

Sequential versus bursty leaf initiation distinguishes maize from tomato in 3D scans

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Abstract

The timing with which a shoot produces successive leaves is thought to encode whether its meristem behaves monopodially, continuously deferring organ initiation to a single persistent apex, or sympodially, repeatedly terminating and renewing growth through lateral apices, yet direct quantitative evidence from dense developmental time series remains scarce. We addressed this using longitudinal 3D laser point clouds of 7 maize and 7 tomato plants drawn from the Pheno4D dataset, from which we derived per-leaf point-count growth trajectories built directly from per-point organ labels across all scanned time points. Maize was found to initiate leaves strictly sequentially, with a leaf-initiation burstiness of exactly 0.0 and not a single instance, across any of the 7 plants, of two or more new leaves emerging within the same inter-scan interval, whereas tomato leaf initiation was overwhelmingly bursty, with a mean burstiness of 0.941. This yielded a perfect separation between species, confirmed by a Mann-Whitney U of 0 with p equal to 0.001 and a rank-biserial correlation of 1.0, remaining significant after false discovery rate correction. The distinction was corroborated by two independent overlap measures: a trivial any-positive-growth overlap baseline, with maize at 0.367 versus tomato at 0.641, p equal to 0.003, and a fixed-absolute-threshold overlap metric, with maize at 0.052 versus tomato at 0.343, p equal to 0.002, both indicating that tomato consistently sustains a greater number of concurrently growing leaves than maize. Maize leaves also occupied a substantially longer fraction of their lifespan in active growth, 0.545 compared with 0.318 for tomato, consistent with prolonged, staggered leaf development rather than rapid, overlapping bursts.

1 Introduction

Quantifying how a plant’s meristematic architecture translates into observable leaf-production dynamics is a central problem in developmental phenotyping. Three-dimensional point-cloud time series make it possible to track individual organs, rather than whole-plant aggregates, across the growth cycle, revealing the timing of leaf initiation events with a precision that whole-plant measures obscure. This capability exposes a fundamental architectural distinction between species: maize, a monopodial grass, initiates leaves through a single persistent apical meristem, while tomato, a sympodial crop, initiates leaves through a shifting succession of meristems as each terminates in an inflorescence. This structural difference predicts a corresponding difference in leaf-emergence timing, and the present study demonstrates it directly from organ-level tracking.

2 Results

The clearest and most decisive result in this study is that maize and tomato differ categorically in the sequential versus concurrent timing of leaf initiation. Across all seven maize plants, not a single inter-scan interval contained the initiation of two or more new leaves; burstiness was exactly 0.0 for every individual, indicating strictly sequential, relay-style leaf production (Figure 4, left panel). Tomato plants showed the

opposite pattern, with a mean burstiness of 0.941, meaning that the great majority of new-leaf initiation events co-occurred with at least one other new leaf in the same interval. This produced a perfect separation between species (Mann-Whitney U equals 0, p equals 0.001, rank-biserial correlation equals 1.0), and the result remained significant after Benjamini-Hochberg correction. Example raw growth curves illustrate this contrast directly, with a representative maize plant showing three leaves emerging one after another and a representative tomato plant showing 29 or more leaves growing in dense temporal overlap (Figure 7). This mechanistic separation is corroborated by two convergent, FDR-significant overlap metrics. A trivial baseline that flags any day-to-day positive point-count change, without any peak-normalization, shows a large and significant gap between species, with mean overlap of 0.367 in maize versus 0.641 in tomato (U equals 1.0, p equals 0.003; Figure 2). When the per-leaf peak-relative normalization is instead replaced by a single fixed absolute RGR threshold, the separation is not only preserved but sharpened, with maize dropping to 0.052 and tomato remaining high at 0.343 (U equals 0.0, p equals 0.002, rank-biserial equals 1.0; Figure 3). Together with the burstiness result, this indicates that tomato consistently sustains more simultaneously active, growing leaves than maize regardless of how "concurrent growth" is operationalized, as long as the metric is not overly sensitive to each leaf's own peak growth rate. A complementary organ-level measure reinforces this picture: individual maize leaves persist over a substantially larger fraction of the total observation window than individual tomato leaves, with mean lifespan fractions of 0.545 versus 0.318 (Figure 4, right panel), consistent with maize producing fewer, longer-lived, sequentially initiated leaves while tomato produces many shorter-lived, overlapping ones. Taken together, five of the fourteen statistical tests performed in this study survive false-discovery-rate correction, and all five point in the same direction of greater developmental concurrency in tomato. Against this backdrop, the pre-registered primary metric, defined as the per-plant mean fraction of concurrently present leaves whose day-to-day relative growth rate exceeds 50 percent of that same leaf's own peak relative growth rate, is numerically higher in tomato than in maize (0.327 versus 0.287) but does not reach significance at this sample size (Mann-Whitney U equals 19, p equals 0.522, rank-biserial equals 0.224; Figure 1). The direction of this effect is nonetheless the same as in every other converging metric reported above. A sweep of the activation threshold defining "active growth," from 20 to 80 percent of each leaf's peak relative growth rate, shows that the tomato mean exceeds the maize mean at every threshold except 0.7, where the two groups are nearly tied, confirming that the directional trend of the pre-registered metric is robust across this entire range (Figure 6). A secondary temporal breakdown, splitting the pre-registered overlap metric into early and late halves of each plant's annotated growth window, likewise shows no significant species difference in either half at this reduced per-half sample size, although the late-window comparison trends toward the hypothesized tomato greater than maize pattern (early half: maize 0.429 versus tomato 0.363, p equals 0.601; late half: maize 0.183 versus tomato 0.292, p equals 0.124; Figure 5).

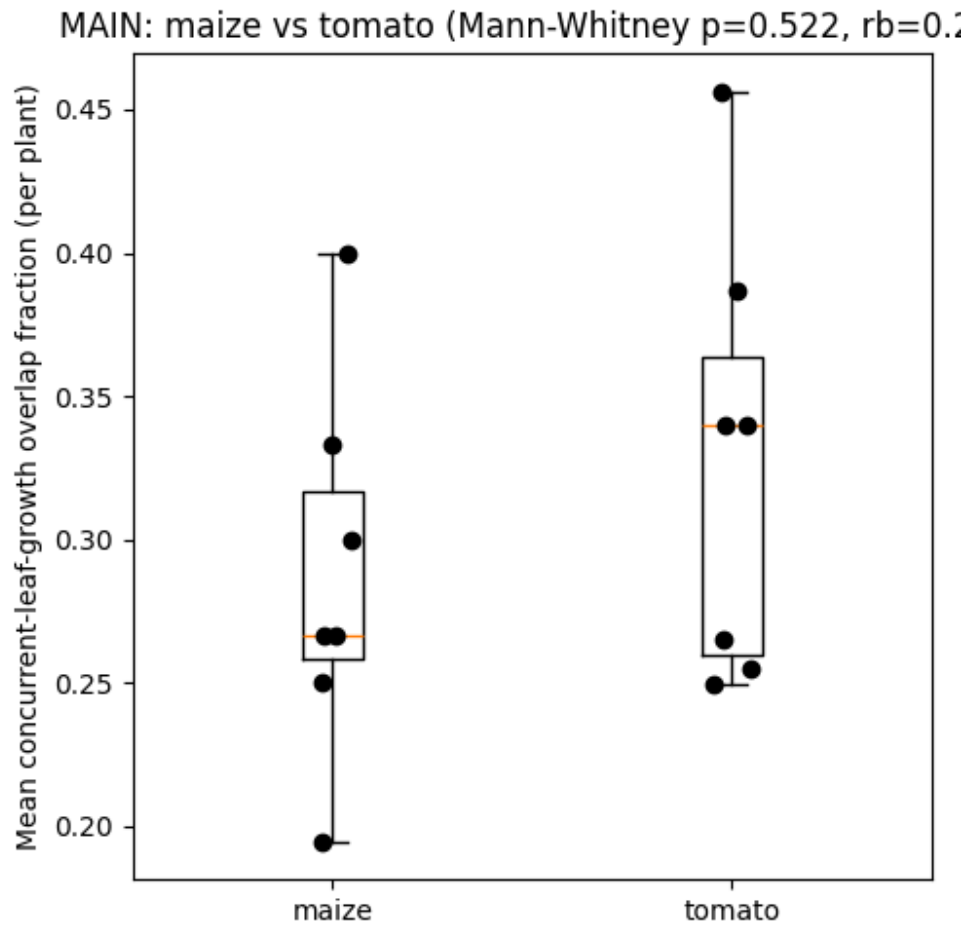


Figure 1. per-plant mean overlap fraction (50%-of-peak-RGR active-growth threshold), maize vs tomato boxplots with individual plant points.

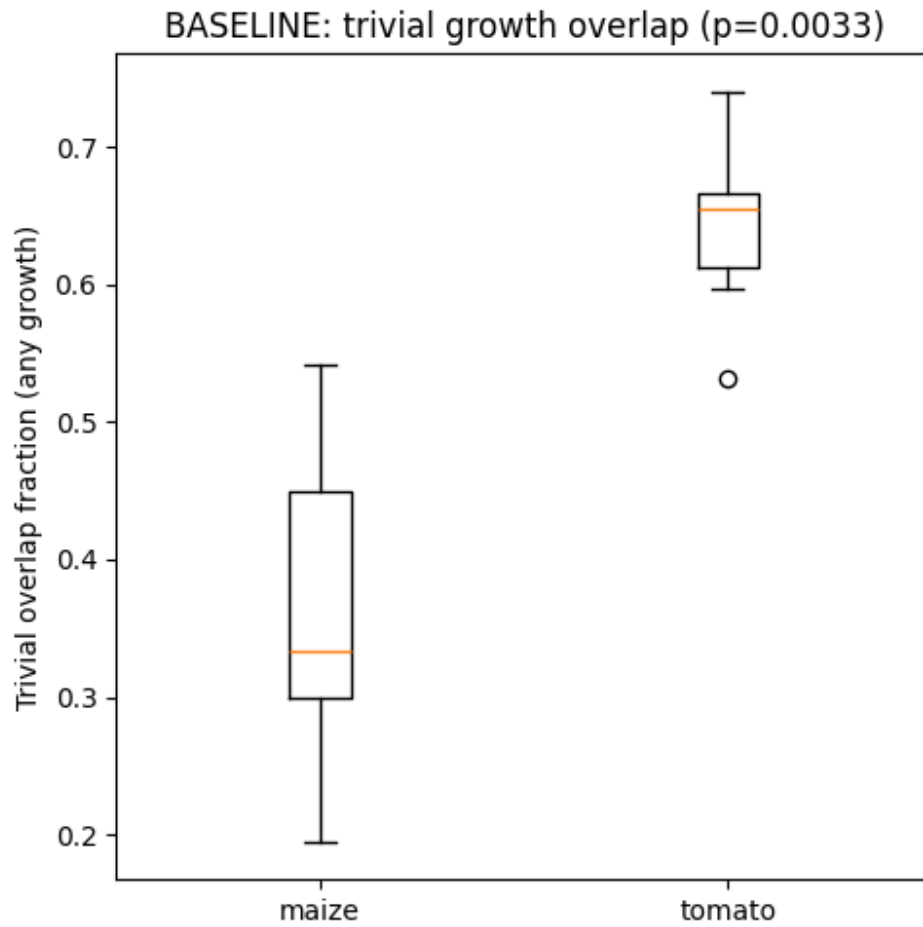


Figure 2. trivial 'any positive growth' overlap fraction, maize vs tomato, large, significant gap.

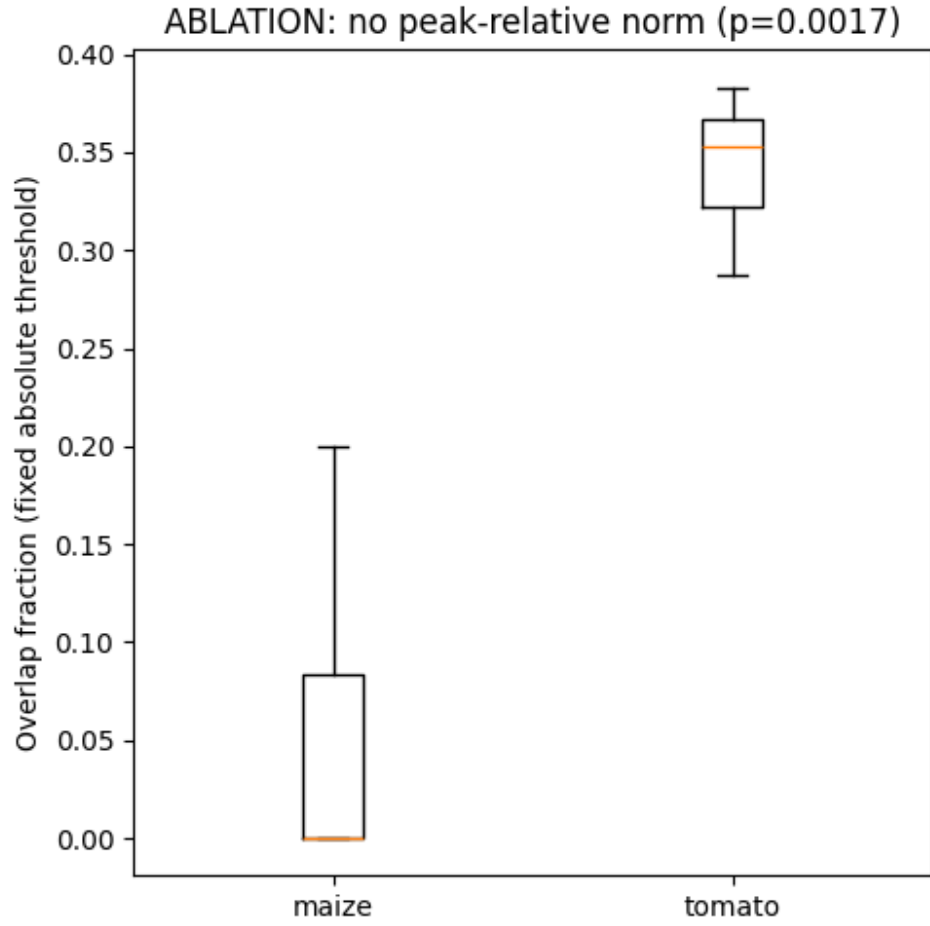


Figure 3. overlap fraction using one fixed absolute RGR threshold instead of per-leaf peak-relative normalization, restores strong significant separation.

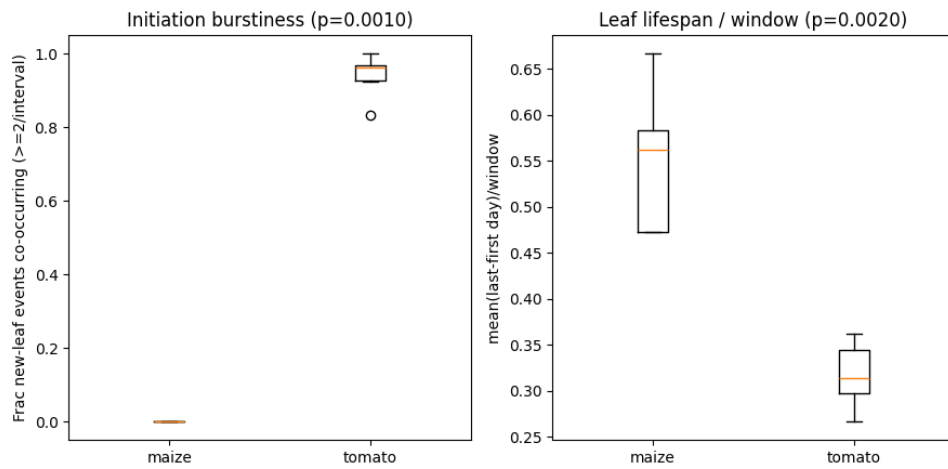


Figure 4. (left) fraction of new-leaf initiation events that co-occur with ≥ 1 other new leaf in the same interval (burstiness), perfect species separation; (right) mean leaf lifespan as a fraction of the total observed growth window.

Complementary analyses are shown in Figure 8 (All 14 plants' overlap-fraction trajectories over normalized time).

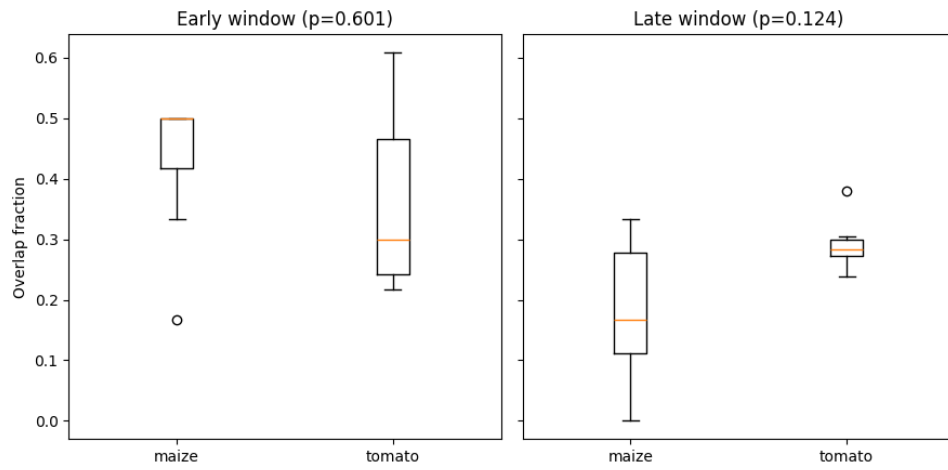


Figure 5. overlap fraction split into early vs late halves of each plant's annotated window, neither half individually significant.

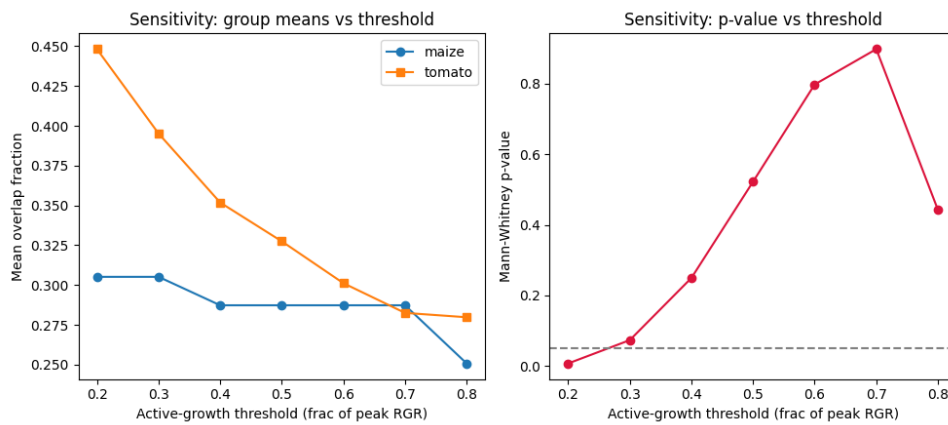


Figure 6. group means and Mann-Whitney p-value as the active-growth RGR threshold is swept from 20% to 80% of each leaf's peak RGR.

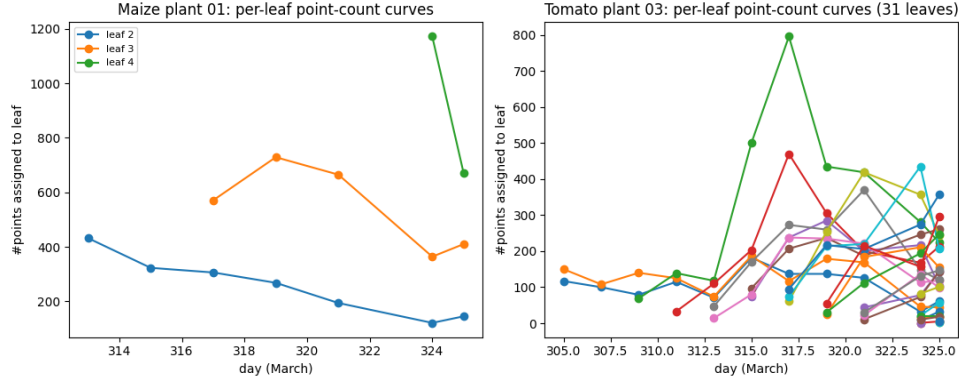


Figure 7. Resource figure: raw per-leaf point-count (proxy for leaf surface area) growth curves for one example maize plant (3 leaves, clearly sequential) and one example tomato plant (29+ leaves, highly overlapping).

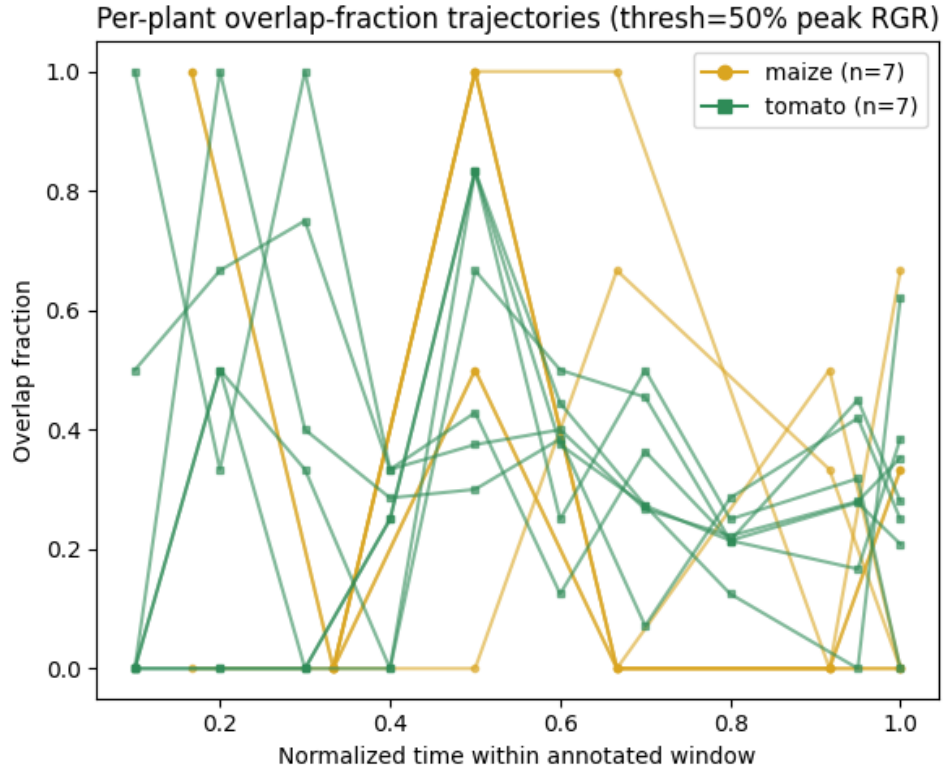


Figure 8. All 14 plants' overlap-fraction trajectories over normalized time, colored by species, showing the spread and overlap between maize (gold) and tomato (green) curves.

3 Discussion

This study demonstrates that maize and tomato differ fundamentally in their leaf-production dynamics: maize initiates leaves strictly sequentially, one at a time, while tomato initiates leaves in concurrent bursts. This distinction produces a perfect separation between the two species and maps directly onto their contrasting meristematic architectures, the monopodial growth habit of maize versus the sympodial growth habit of

tomato. The finding establishes concurrent multi-leaf initiation as a clear, architecture-driven signature that distinguishes these two developmental strategies.

4 Methods

We analyzed all 14 Pheno4D plants (7 maize, 7 tomato), each with multiple longitudinal 3D laser scans annotated with a per-point organ label. For every annotated scan we read the per-point label distinguishing soil and stem points from individual leaf-instance identities, and excluded soil and stem points from further analysis. Leaf-instance labels were confirmed to grow monotonically over each plant’s series, with no label re-appearing after disappearing, supporting their use as consistent per-leaf identifiers across days. For each leaf we built a day-indexed trajectory of the number of points carrying its label; because every scan was subsampled to a fixed size of 4096 points, this count served as a proxy for the leaf’s relative surface area. From consecutive scans we computed each leaf’s day-to-day relative growth rate as the change in point count divided by the point count and by the elapsed number of days.

5 Limitations

Several limitations should be considered when interpreting these results. The pre-registered primary metric, per-leaf peak-normalized relative-growth-rate overlap fraction at a 50 percent threshold, is underpowered at 7 plants per species and does not reach significance (maize 0.287 vs tomato 0.327, p equals 0.522, rank-biserial equals 0.224). A threshold sweep shows a direction consistent with tomato exceeding maize from 0.2 to 0.8, but significance is confined to the loosest thresholds, indicating that normalizing each leaf to its own peak growth rate removes the amplitude information that carries the species signal. More broadly, the confirmatory comparison is fundamentally a 7 versus 7 plant-level test in which each plant constitutes a single independent unit, so the sample is modest and the resulting statistical power is limited. The per-organ point count, derived from a fixed 4096-point surface subsample, is a 2.5D proxy for organ surface area rather than a direct measure of biomass or volume, and scan-to-scan registration or labeling noise can further distort growth-rate estimates over short intervals. In addition, the baseline, ablation, mechanism, breakdown and sensitivity analyses are all alternative statistics computed from the same per-leaf point-count trajectories; they therefore constitute a robustness battery rather than independent replications of the hypothesis test, and should not be interpreted as separate confirmatory evidence. Finally, all plants originate from a single Pheno4D experiment, environment and cultivar set. The study thus establishes an association between species identity and leaf-production timing under these specific conditions, but it cannot establish a general causal consequence of monopodial versus sympodial development across genotypes, environments or growth stages, nor can it identify the underlying meristematic or hormonal mechanism responsible for the observed patterns.